Check-in 1: Topic + Theory of Change Luke Keller

1. Topic and Motivation:

Tuberculosis (TB) remains one of the leading causes of infectious disease mortality worldwide, and Indonesia bears one of the heaviest burdens globally. Despite substantial investments in diagnosis and treatment, Indonesia continues to experience high TB incidence and mortality rates, driven in part by widespread underreporting and delayed case notification. Many individuals with TB, especially those receiving care in private facilities, are never reported to the National Tuberculosis Program (NTP), limiting the government's ability to monitor transmission and deliver timely treatment.

This problem disproportionately affects low-income populations, working-age adults, and individuals with comorbidities such as HIV and diabetes. These groups face higher risks of infection, worse treatment outcomes, and greater financial hardship. Underreporting also weakens public health surveillance systems, allowing TB transmission to continue unchecked and increasing the risk of drug-resistant TB.

Currently, Indonesia relies on mandatory reporting policies, digital platforms such as SITT and mobile applications, and public-private partnerships to improve case notification; however, implementation remains uneven, particularly in private clinics and hospitals. Providers face high workloads, limited incentives, technical barriers, and concerns about patient confidentiality. New evidence is needed to identify which interventions can effectively improve reporting behavior and strengthen coordination between private providers and public health authorities. Without rigorous evaluation, policymakers cannot determine whether investments in digital systems, incentives, or training are producing meaningful health improvements. This project aims to fill this gap by evaluating strategies to improve TB notification and follow-up in Indonesia's mixed public-private healthcare system.

2. Research Question:

Does providing technical assistance and digital reminder support to private primary care clinics increase tuberculosis case notification and reduce reporting delays in District X of Indonesia?

Treatment (X):
Access to the low-friction support package.

Outcomes (Y):
Monthly TB notifications per clinic
% of diagnosed cases reported

Population:
Private primary care clinics in District X.

3. Market Failures and Logic:

Private clinics diagnose and treat TB patients, but the social benefits of reporting, like improved disease surveillance, reduced transmission, and better resource allocation, are more captured by the public health system rather than by individual providers. As a result, providers do not internalize the full benefits of reporting, leading to systematic underreporting (negative externality?). They would likely prioritize immediate patient care and revenue-generating activities over unpaid administrative tasks.

In addition, information asymmetry might exist between private providers and public health authorities. Many providers seem to lack clear knowledge about reporting requirements/processes, digital systems, and follow-up procedures, potentially as well as the why behind the reporting.

Overall, because reporting generates few private benefits and imposes nontrivial costs, the socially optimal level of notification does not occur naturally. Without intervention, private clinics underinvest in reporting despite its high public value.

Logic diagram included at the end

Key Assumptions:
Technical and informational barriers are a primary cause of underreporting.
Providers are willing to comply when reporting costs are reduced.
Digital systems function reliably.
Public health agencies respond more effectively when cases are reported more accurately.

Mechanisms Linking Intervention to Outcomes:
Lowering administrative burden reduces cost of task.
Training improves understanding of procedures.
Reminders increase attention and compliance.
Provider confidence improves with support.

Counterfactual:
Without the intervention, as observed in the control group, private clinics would continue to face high administrative and technical barriers to reporting, leading to persistent underreporting, delayed case notification, and weakened TB surveillance and control.

4. Initial Literature Scan:

I. Tuberculosis in Indonesia: challenges and future directions
Saktiawati, Antonia Morita Iswari et al.
The Lancet Respiratory Medicine, Volume 13, Issue 8, 669 - 671

This article analyzes Indonesia's tuberculosis burden and evaluates recent national strategies to make progress toward WHO TB targets. It synthesizes survey data, policy initiatives, and health system reforms to identify gaps in prevention, diagnosis, financing, and treatment coverage. The paper's main contribution is a framework of system-level interventions, linking financing, digital health, workforce development, and social protection to strengthen tuberculosis control.

II. Suryanti, Ahmed IA. Recent Review Tuberculosis in Indonesia: Burden and the Challenge of Under-Reporting. Iran J Public Health. 2025 Feb;54(2):445-446. PMID: 40225262; PMCID: PMC11992917.

This article provides a comprehensive overview of some of the social and economic burdens of tuberculosis in Indonesia, emphasizing underreporting as a central barrier to disease control. It demonstrates how gaps in case notification, particularly in the private sector, undermine monitoring, treatment, and prevention efforts. It also highlights key structural, behavioral, and technological constraints affecting reporting systems. It points at evaluating interventions aimed at improving TB case detection and notification.

III. Lestari T, Fuady A, Yani FF, Putra IWGAE, Pradipta IS, Chaidir L, Handayani D, Fitriangga A, Loprang MR, Pambudi I, Ruslami R, Probandari A. The development of the national tuberculosis research priority in Indonesia: A comprehensive mixed-method approach. PLoS One. 2023 Feb 9;18(2):e0281591. doi: 10.1371/journal.pone.0281591. PMID: 36758064; PMCID: PMC9910756.

This paper seems useful. It develops Indonesia's first comprehensive, stakeholder-driven national tuberculosis research agenda using a mixed-methods approach that combines literature review, nationwide surveys, and expert Delphi panels. It identifies gaps between existing research, frontline needs, and national TB elimination targets, highlighting areas like missing case detection, drug-resistant TB, contact investigation, and preventive therapy. The paper's main contribution is an attempt to align future TB research with policy and implementation priorities to strengthen the translation of research into effective public health action.

What Gaps Does This Project Fill?

While existing studies document substantial underreporting of tuberculosis cases in Indonesia and emphasize the importance of early detection and surveillance, there is limited causal evidence on whether low-cost, clinic-level administrative support, or something else, can improve reporting behavior in the private sector. This project will address this gap by evaluating a feasible, scalable intervention that reduces reporting barriers and examines its short-term effects on case notification and timeliness.

Interv. → Interm. Outcomes → Behavior change → Outcomes → Human Result

Interv.

Interm. Outcomes

- Reduced time + effort needed to notify cases
- Increased private knowledge of reporting procedures
- Increased salience of reporting obligation

Behavior change

- Higher probability that diagnosed cases are reported

Outcomes

- Better contact tracing, reducing transmission
- Better linking diagnosed patients with follow up measures
- Facilitates efficient resource allocation to high burden areas
- Informs future policy decisions with more robust data

Human Result

- Less people get and die of TB in Indonesia